

Introducción al Análisis Matemático

Recuperatorio del 1º parcial - 18/06/2024

Tema π

Nombre, apellido y carrera
Scheuma Sofia - Ingeniería en Alimentos

1º	2º	3º	4º	Condición
B	B/B	B	B	¡No! Faltaba algo

FINAL

Ejercicio 1: Dadas las funciones $[f(x) = |x+4|+2]$ y $[g(x) = x^2+6\cdot x+10]$

Calcular $\{x \in \mathbb{R} / g(x) > f(x)\}$

Ejercicio 2: Sea $f(x) = \log_{\frac{1}{4}}\left(\frac{2\cdot x-2}{x^2-x-12}\right)$.

1. Calcular $\text{dom}(f)$.

B

2. Calcular $C^+(f)$, $C^-(f)$ y $C^0(f)$.

Ejercicio 3: Calcular las asíntotas horizontales y verticales de la función

$$f(x) = \frac{\sin(x-3)}{\frac{1}{2}x - \frac{9}{2} + |x-6|}$$

Ejercicio 4: Calcular $\lim_{x \rightarrow -\infty} \left(\frac{x+3}{x+1}\right)^{\sqrt{x^2+1}}$

Ejercicio 1

$$g(x) > f(x)$$

$$x^2 + 6x + 10 > |x+4| + 2$$

$$|x+4| \begin{cases} x+4 & \text{si } x \geq -4 \\ -x-4 & \text{si } x < -4 \end{cases}$$

Caso ① $x \in [-4; +\infty)$

$$x^2 + 6x + 10 > |x+4| + 2$$

$$x^2 + 6x + 10 > x+4+2$$

$$x^2 + 6x + 10 > x+6$$

$$x^2 + 6x - x + 10 - 6 > 0$$

$$x^2 + 5x + 4 > 0$$

CUADRÁTICA
BUSCARCER

$$a > 0 \quad \begin{array}{c} -4 \quad -1 \\ \text{C}^0 = \{-4; -1\} \end{array} \quad S = (-\infty; -4) \cup (-1; +\infty) \cap x \in [-4; +\infty)$$

$$S_1 = (-1; +\infty)$$

Caso ② $x \in (-\infty; -4)$

$$x^2 + 6x + 10 > |x+4| + 2$$

$$x^2 + 6x + 10 > -x-4+2$$

$$x^2 + 6x + 10 > -x-2$$

$$x^2 + 6x + x + 10 + 2 > 0$$

$$x^2 + 7x + 12 > 0$$

$$a > 0 \quad \begin{array}{c} -4 \quad -3 \\ \text{C}^0 = \{-4; -3\} \end{array} \quad S = (-\infty; -4) \cup (-3; +\infty) \cap x \in (-\infty; -4)$$

$$S_2 = (-\infty; -4)$$

$$S_1 \cup S_2 = S = (-\infty; -4) \cup (-1; +\infty)$$

Ejercicio 2

$$f(x) = \log_{\frac{1}{4}} \left(\frac{2x-2}{x^2-x-12} \right)$$

① Para el Dominio

$$\frac{2x-2}{x^2-x-12} > 0 \rightarrow \text{porque el argumento de un log debe ser } > 0$$

$$x^2-x-12 \neq 0 \rightarrow \text{porque el denominador no puede ser } = 0$$

$$a=1 \quad b=-1 \quad c=-12$$

$$\frac{-(-1) \pm \sqrt{(-1)^2 - 4 \cdot 1 \cdot (-12)}}{2 \cdot 1} \begin{cases} x_1 = 4 \\ x_2 = -3 \end{cases}$$

$$\text{Caso 1 } \frac{+}{+}$$

$$2x-2 > 0 \quad y \quad x^2-x-12 > 0$$

$$a > 0$$

$$C^0 = \{-3, 4\}$$

$$x \in \mathbb{R} - \{-3, 4\}$$

$$x > 1 \quad \cap \quad x \in (-\infty, -3) \cup (4, +\infty)$$

$$S_0 = (4, +\infty)$$

$$\text{Caso 2 } \frac{-}{-}$$

$$2x-2 < 0 \quad y \quad x^2-x-12 < 0$$

$$a > 0$$

$$C^0 = \{-3, 4\}$$

$$x < 1 \quad \cap \quad x \in (-3, 4)$$

$$S_2 = (-3, 1)$$

$$\text{Entonces, el } \text{Dom}(f) = (-3, 1) \cup (4, +\infty)$$

② Para calcular los ceros de la función

$$\log_{\frac{1}{4}} \left(\frac{2x-2}{x^2-x-12} \right) = 0$$

por definición de logaritmo

$$\frac{2x-2}{x^2-x-12} = \left(\frac{1}{4} \right)^0$$

$$\frac{2x-2}{x^2-x-12} = 1$$

$$2x-2 = x^2-x-12$$

$$-x^2 + 3x + 10 = 0$$

$$\frac{-(-3) \pm \sqrt{3^2 - 4 \cdot (-1) \cdot 10}}{2 \cdot (-1)}$$

$$x_1 = -2 \quad x_2 = 5$$

$$\text{Intersección con el Dom} \quad C^0 = \{-2, 5\}$$

$$\log_{\frac{1}{4}} \left(\frac{2x-2}{x^2-x-12} \right) > 0$$

Aplico def. de logaritmo y como $0 < a < 1$ invierto el signo.

$$\frac{2x-2}{x^2-x-12} < \left(\frac{1}{4}\right)^0$$

Case ① $\frac{-}{+}$

$$\frac{2x-2}{x^2-x-12} < 1$$

$-x^2 + 3x + 10 < 0$ y $x^2 - x - 12 > 0$
 $a < 1$  $a > 1$ 
 $C^0 = \{-2; 5\}$ $C^0 = \{-3; 4\}$

$a < 1$

$C^0 = \{-2; 5\}$

$a > 1$

$C^0 = \{-3, 4\}$



$$C^0 = \{-2; 5\}$$

$$C^0 = \{-3; 4\}$$

$$\frac{2x-2}{x^2-x-12} - 1 < 0$$

$$\frac{-x^2 + 3x + 10}{x^2 - x - 12} < 0 \quad \begin{array}{l} \text{separa} \\ \text{em dois} \\ \text{casos} \end{array} \rightarrow$$

$$x \in (-\infty, -2) \cup (5, +\infty) \cap x \in (-\infty, -3) \cup (4, +\infty)$$

$$S = (-\infty; -3) \cup (5; +\infty)$$

Case (2) $\frac{+}{-}$

$$-x^2 + 3x + 10 > 0 \quad \text{y} \quad x^2 - x - 12 < 0$$

$a < 1$

$C^0 = \{-2, 5\}$

A hand-drawn graph of a parabola opening downwards. The x-axis is represented by a horizontal line. Two points are marked on the x-axis with dots and labeled '-2' and '5'. The parabola passes through these two points, which are its x-intercepts. The vertex of the parabola is located above the x-axis, between the points -2 and 5.

$a > 1$

$C^0 = \{-3, 4\}$



$$C^0 = \{-2, 5\}$$

$$C^0 = \{-3, 4\}$$

$$x \in (-2; 5) \quad y \quad x \in (-3; 4)$$

$$(S_0 \cup S_2) \cap \text{Dom } F = C^+ = (-2; 1) \cup (5; +\infty)$$

$$(-\infty, -3) \cup (-2, 4) \cup (5, +\infty) \quad \cap \quad (-3, 1) \cup (4, +\infty)$$

Number line for the second inequality: $x < 4$. The number line is marked from -3 to 5. An open circle is at 4, and the region to the left of 4 is shaded green.

Como Dom $f = (-3, 1) \cup (4, +\infty)$

$$y \quad e^+ = (-2; 1) \cup (5; +\infty)$$

y $C^+ = (-2; 1) \cup (5; +\infty)$
 En términos $C^- = (-3; -2) \cup (4; 5) \rightarrow$ porque C^- es el domf
 sim C^+ y sim los C^0

Ejercicio 3

$$f(x) = \frac{\sin(x-3)}{\frac{1}{2}x - \frac{9}{2} + |x-6|}$$

$$|x-6| = \begin{cases} x-6 & \text{si } x \geq 6 \\ -x+6 & \text{si } x < 6 \end{cases}$$

Para el dominio

$$\frac{1}{2}x - \frac{9}{2} + |x-6| \neq 0$$

$$x \in [6; +\infty)$$

$$\frac{1}{2}x - \frac{9}{2} + x - 6 \neq 0$$

$$\left[\frac{3}{2}x - \frac{21}{2} \neq 0 \right] \times 2 \begin{matrix} \text{multiplico} \\ \text{por 2} \\ \text{ambos miembros} \end{matrix}$$

$$3x - 21 \neq 0$$

$$x \neq \frac{21}{3}$$

$$x \neq 7$$

$$x \in (-\infty; 6)$$

$$\frac{1}{2}x - \frac{9}{2} - x + 6 \neq 0$$

$$\left[-\frac{1}{2}x + \frac{3}{2} \neq 0 \right]$$

$$-x + 3 \neq 0$$

$$x \neq \frac{-3}{-1}$$

$$x \neq 3$$

Dom $f(x) = \mathbb{R} - \{3; 7\}$ Posibles A.V.

$$\lim_{x \rightarrow 3^+} \frac{\sin(x-3)}{\frac{1}{2}x - \frac{9}{2} - x + 6} = \lim_{x \rightarrow 3^+} \frac{\sin(x-3)}{\frac{1}{2}x + \frac{3}{2}} \text{ IND } \frac{0}{0} \begin{matrix} \text{Multiplico } \times 2 \\ \text{ambos miembros} \end{matrix}$$

$$\lim_{x \rightarrow 3^+} \frac{-2 \cdot \sin(x-3)}{+1x - 3} = \lim_{x \rightarrow 3^+} -2 \cdot \frac{\sin(x-3)}{x-3} \cdot 1 = -2$$

$$\lim_{x \rightarrow 3^-} \frac{\sin(x-3)}{\frac{1}{2}x - \frac{9}{2} - x + 6} = \lim_{x \rightarrow 3^-} \frac{\sin(x-3)}{-\frac{1}{2}x + \frac{3}{2}} = \lim_{x \rightarrow 3^-} -2 \cdot \frac{\sin(x-3)}{x-3} \cdot 1 = -2$$

$x=3$ NO es A.V. de $f(x)$

$$\lim_{x \rightarrow 7^+} \frac{\sin(x-3)}{\frac{1}{2}x - \frac{9}{2} + x - 6} = \lim_{x \rightarrow 7^+} \frac{\sin(x-3)}{\frac{3}{2}x - \frac{21}{2}} \quad \begin{matrix} \text{sem}(4) \neq 0 \\ \frac{N}{D} = \frac{\infty}{\infty} \end{matrix}$$

$$\lim_{x \rightarrow 7^-} \frac{\sin(x-3)}{\frac{3}{2}x - \frac{21}{2}} = \frac{N}{D} = \frac{\infty}{\infty} \quad \left\{ x=7 \text{ es A.V. de } f(x) \right\}$$

Evaluación en $x=6$, punto de corte

$$\lim_{x \rightarrow 6^-} \frac{\sin(x-3)}{\frac{1}{2}x - \frac{9}{2} - x + 6} = \lim_{x \rightarrow 6^-} \frac{\sin(x-3)}{-\frac{1}{2}x + \frac{3}{2}} = \frac{\sin(3)}{-\frac{3}{2}}$$

$$\lim_{x \rightarrow 6^+} \frac{\sin(x-3)}{\frac{1}{2}x - \frac{9}{2} - x + 6} = \lim_{x \rightarrow 6^+} \frac{\sin(x-3)}{\frac{3}{2}x - \frac{21}{2}} = \frac{\sin(3)}{-\frac{3}{2}}$$

$x=6$ NO es A.V. de $f(x)$

Ahora evaluación en $+\infty$ y $-\infty$ para buscar A.H.

$$\lim_{x \rightarrow +\infty} \frac{\sin(x-3)}{\frac{1}{2}x - \frac{9}{2} - x + 6} = \lim_{x \rightarrow +\infty} \frac{\sin(x-3)}{\frac{3}{2}x - \frac{21}{2}} = \lim_{x \rightarrow +\infty} \frac{1}{\frac{3}{2}x - \frac{21}{2}} \quad \begin{matrix} \text{ACOTADO} \\ \sin(x-3) = 0 \end{matrix}$$

$$\lim_{x \rightarrow -\infty} \frac{\sin(x-3)}{\frac{1}{2}x - \frac{9}{2} - x + 6} = \lim_{x \rightarrow -\infty} \frac{\sin(x-3)}{-\frac{1}{2}x + \frac{3}{2}} = \lim_{x \rightarrow -\infty} \frac{1}{-\frac{1}{2}x + \frac{3}{2}} \quad \begin{matrix} \text{ACOTADO} \\ \sin(x-3) = 0 \end{matrix}$$

$y=0$ es A.H. de $f(x)$

Ejercicio 4

$$\lim_{x \rightarrow -\infty} \left(\frac{x+3}{x+1} \right)^{\sqrt{x^2+1}} = \lim_{x \rightarrow -\infty} \left[\frac{x \cdot \left(1 + \frac{3}{x}\right)}{x \cdot \left(1 + \frac{1}{x}\right)} \right]^{\sqrt{x^2+1}} = \text{IND } 1^\infty$$

$$\begin{aligned} \lim_{x \rightarrow -\infty} \left(1 + \frac{x+3}{x+1} - 1 \right)^{\sqrt{x^2+1}} &= \lim_{x \rightarrow -\infty} \left(1 + \frac{2}{x+1} \right)^{\sqrt{x^2+1}} \\ &= \lim_{x \rightarrow -\infty} \left(1 + \frac{1}{\frac{x+1}{2}} \right)^{\frac{x+1}{2} \cdot \frac{\sqrt{x^2+1}}{\frac{x+1}{2}}} = \lim_{x \rightarrow -\infty} \left[\left(1 + \frac{1}{\frac{x+1}{2}} \right)^{\frac{x+1}{2}} \right]^{\frac{\sqrt{x^2+1}}{\frac{x+1}{2}}} \end{aligned}$$

$$\lim_{x \rightarrow -\infty} e^{\frac{\sqrt{x^2+1}}{\frac{x+1}{2}}} = \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2+1}}{\frac{x+1}{2}} = \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2} \cdot \sqrt{1+\frac{1}{x^2}}}{\frac{x}{2} \cdot \left(1 + \frac{1}{x}\right)} = \lim_{x \rightarrow -\infty} \frac{x \cdot \sqrt{1+\frac{1}{x^2}} \cdot 2}{x \cdot \left(1 + \frac{1}{x}\right)} = 2$$

$$\lim_{x \rightarrow -\infty} \frac{x \cdot \sqrt{1+\frac{1}{x^2}} \cdot 2}{x \cdot \left(1 + \frac{1}{x}\right)} = 2$$

$$\lim_{x \rightarrow -\infty} \frac{x \cdot \sqrt{1+\frac{1}{x^2}} \cdot 2}{x \cdot \left(1 + \frac{1}{x}\right)} = 2$$

$$\lim_{x \rightarrow -\infty} = e^2$$